

Hongwei (Henry) Zhou

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EDUCATION

2019-2025 **University of California, Santa Cruz**

PH.D. AND M.S. IN COMPUTATIONAL MEDIA

- **M.S. Thesis:** Islands and Bridges of Language: Bio-Inspired Structural Analysis of Language Embedding Data
- **Ph.D. Thesis:** Theorize, Interpret, and Design Video Game Systems through Anti-Essentialist Philosophy

2015-2019 **New York University, Tandon School of Engineering**

B.S. IN COMPUTER SCIENCE, MINOR IN GAME ENGINEERING

POSITIONS HELD

2025-2026 **Postdoctoral Fellowship**

SCHOOL OF COMPUTER SCIENCE. CARLETON UNIVERSITY

Conceptualized a theoretical approach for a game design & deployment that synthesizes social criticality with game making. Mentored students in integrating research around issues of diversity and inclusivity in academia with their own game design practices. Created and experimented with a game-based learning method inspired by critical pedagogy and transformative learning.

2025 **Graduate Student Instructor**

COMPUTATIONAL MEDIA. UNIVERSITY OF CALIFORNIA, SANTA CRUZ

Designed and taught "Video Games and Computational Culture" based on dissertation work. Connected game design principles to relevant humanistic studies of media, games and technoculture.

PUBLICATIONS

BOOK CHAPTERS

- 2027 **Zhou, H.**, Wang, Y., Almeda, S., Wang, T., Sun, Y., Roemmele, M., Jennings, N., Li, I., Chung, J., Kreminski, M. Negotiating Narrative Agency in LLM-Mediated Interactive Visual Novels. *Many, Many Machine Agencies*. [Submitted]

JOURNAL ARTICLES

- 2027 Szozda, A., **Zhou, H.**, Quinland, C., Villeneuve, J., Zhang, A., Salmaso, N., Thomson, R., Melcer, E., Chiasson, S. A Systematic Review of Technological and Game-Based Equity, Diversity, and Inclusion Interventions in Higher Education. [In-Progress, Journal Undecided]
- 2025 **Zhou, H.**, Meng, F., K., & Wardrip-Fruin, N. (2025, April). More than Interactivity: Designing a Critical AI Game beyond Ludo-Centrism. *Humanities*, 14(4), 88.
- 2024 **Zhou, H.** (2024, December). Infrastructure of Agency: An Anti-essentialist and Post-humanist Framework for Video Game Agency. *Journal of Games Criticism*, 6(A), 1-22.

CONFERENCE PAPERS

- 2027 Almeda, S., Wang, T., **Zhou, H.**, Sun, Y., Roemmele, M., Wang, Y., Jennings, N., Li, I., Chung, J. & Kreminski, M. *Whose Line Is It Anyway? Exploring LLM-Mediated Interactive Narrative with Dramamancer*. Conference on Human Factors in Computing Systems (CHI) 2027 [Submitted]
- 2024 **Zhou, H.**, Wardrip-Fruin, N., & Mateas, M. (2024, September). *Corporeal Capture: The Rhetoric of Boundaries in Procedurality*. In Proceedings of DiGRA 2024 Conference: Playgrounds.
- 2024 **Zhou, H.**, Zhu, J., Mateas, M., & Wardrip-Fruin, N. (2024, May). *The Eyes, the Hands and the Brain: What can Text-to-Image Models Offer for Game Design and Visual Creativity?*. In Proceedings of the 19th International Conference on the Foundations of Digital Games (FDG) (pp. 1-13).
- 2022 **Zhou, H.**, Gonzalez, K., Altice, N., Wardrip-Fruin, N., & Forbes, A. G. (2022, January). *On the Maintenance of Meaning: A Deleuzian View on Proceduralism*. In Proceedings of DiGRA 2022 Conference: Bringing Worlds Together.
- 2018 **Zhou, H.**, Gong, Y., Mugrai, L., Khalifa, A., Nealen, A., & Togelius, J. (2018, August). *A hybrid search agent in pommernan*. In Proceedings of the 13th international conference on the foundations of digital games (pp. 1-4).
- 2018 Gopstein, D., **Zhou, H. H.**, Frankl, P., & Cappos, J. (2018, May). *Prevalence of confusing code in software projects: Atoms of confusion in the wild*. In Proceedings of the 15th international conference on mining software repositories (pp. 281-291).

WORKSHOP PAPERS/POSTERS/LATE-BREAKING

- 2026 Zhang, A., **Zhou, H.**, Hellemans, K., & Melcer, E. (2026, August). *Addressing Anxiety Through Games: A Review of Therapeutic Design Strategies*. Foundations of Digital Games (FDG) 2026.
- 2026 Villeneuve, J., **Zhou, H.**, & Melcer, E. (2026, August). *Serious Games for Cognitive Bias Mitigation: A Review of Learning Mechanisms and Design Approaches*. Foundations of Digital Games (FDG) 2026.
- 2022 **Zhou, H.**, & Forbes, A. G. (2022, October). *Data Feel: Exploring Visual Effects in Video Games to Support Sense-making Tasks*. In 2022 IEEE 7th Workshop on Visualization for the Digital Humanities (VIS4DH) (pp. 6-12). IEEE.
- 2020 **Zhou, H. H.**, Elek, O., Anand, P., & Forbes, A. G. (2020, October). *Bio-inspired structure identification in language embeddings*. In 2020 IEEE 5th Workshop on Visualization for the Digital Humanities (VIS4DH) (pp. 7-13). IEEE.

OTHER PUBLICATIONS

- 2026 **Zhou, H.**. *Book Review of The Impossible Reversal: A History of How We Play by Peter D. McDonald*. Critical Inquiry. [In-Progress]

FUNDRAISING

COMPLETED

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|-----------|---|----------|
| 2024-2025 | Coha-Gunderson Prize in Speculative Futures for Exhibiting and Workshopping
Dissertation Game <i>Sea of Paint</i> , The Humanities Institute | \$ 2,000 |
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PRESENTATIONS & TALKS

INVITED TALKS & PANELS & SYMPOSIUMS

- 2026 **Zhou, H.** *Corporeal Capture: Object Orientation as Boundary-drawing Practices in Video Games*. "Storage Units: Video Games as Containers". Society for Social Studies of Science (4S) 2026, Toronto, Canada
- 2026 **Zhou, H.**, Quinland, C., Villeneuve, J., Zhang, A., Melcer, E., Chiasson, S. *Initial Lessons of Creating Educational Games in EDI Initiatives*. Symposium Talk: Canadian Association of Physics (CAP) 2026, Ottawa, Canada.

CONFERENCE

- 2027 **Zhou, H.**, Bashandy, H. *A Man Chooses, the Crowd Obeys: On Scalar Knowability of the Crowd in Video Games*. Extended Abstract: Society for Cinema and Media Studies (SCMS) 2027, Montréal, Canada. [Submitted]
- 2026 **Zhou, H.**, Melcer, E., Chiasson, S. *Game as Inquiry: A Diffractive and Experimental Approach for Games with a Message*. Extended Abstract: Digital Games Research Association (DiGRA) 2026, Maynooth, Ireland.
- 2026 **Zhou, H.** *Differences in Repetition: Habituation of Game Rules in Painting Games*. Extended Abstract: Canadian Game Studies Association (CGSA) 2026, Montreal, Canada.
- 2025 **Zhou, H.**, Gonzalez, K. *Administer like a Scientist: The Play Aesthetics of Emergence in Agent-based Modeling*. Extended Abstract: Society for Literature, Science & the Arts (SLSA) 2025, Oregon, USA.
- 2025 **Zhou, H.** *Ludo-Computationalism: How Computation Shapes Game Scholarship*. Extended Abstract: Digital Games Research Association (DiGRA) 2025 Conference, Valleta, Malta.
- 2023 **Zhou, H.** *Reframing Posthuman Agency through the Concept of Speed*. Extended Abstract: Digital Games Research Association (DiGRA) 2023 Conference, Sevilla, Spain.

DEMOS

- 2026 **Zhou, H.**, Meng, F., Kosolapova, K. (2025, April). *Sea of Paint: A Critical AI, Text-to-Image Narrative Game Without Text-to-Image Model*. Foundations of Digital Games 2025.

PUBLIC WORKS & EXHIBITIONS

- 2025 *Sea of Paint*. Hongwei Zhou, Katherine Kosolapova, Kyle Gonzalez, Fandi Meng. Digital Game. "Amending Worlds". Santa Cruz Museum of Art & History. Santa Cruz. California. USA. June 5th-15th.
- 2021 *Four Gentlemen*. Henry Zhou. Printed Paintings. "In These Uncertain Times". Santa Cruz Museum of Art & History. Santa Cruz. California. USA. March 12th - May 2nd.

TEACHING

INSTRUCTOR ON RECORD

Su2025 CPM 180, Video Games and Computational Culture

University of
California,
Santa Cruz

TEACHING ASSISTANT IN UNIVERSITY OF CALIFORNIA, SANTA CRUZ

Sp2025 **AGPM 172**, Game Design Studio III
 W2025 **CMPM 146**, Game AI
 Su2024 **CMPM 80K**, Foundations of Video Game Design
 Sp2024 **CMPM 80I**, Introduction to Human Computer Interaction
 W2024 **CMPM 80J**, Games as Technical Culture
 F2023 **CMPM 25**, Introduction to 3D Modeling
 Su2023 **CMPM 80K**, Foundations of Video Game Design
 Sp2023 **CMPM 151**, Algorithmic Music for Games
 W2023 **CMPM 176**, Game Systems
 F2022 **CMPM 25**, Introduction to 3D Modeling
 Sp2022 **CMPM 80K**, Foundations of Video Game Design
 W2022 **CMPM 176**, Game Systems
 F2021 **CMPM 25**, Introduction to 3D Modeling
 Sp2021 **CMPM 172**, Game Design Studio III
 W2021 **CMPM 171**, Game Design Studio II
 F2020 **CMPM 170**, Game Design Studio I
 Su2020 **CMPM 146**, Game AI
 W2020 **CMPM 35**, Data Structures for Interactive Media
 F2019 **CMPM 179/ARTG 179**, Game Design Practicum

INVITED LECTURE

F2025 **COMP 4900D/5900L**, Games for Social Good

F2019 **CMPM 179/ARTG 179**, Game Design Practicum

*Carleton
University
University of
California,
Santa Cruz*

OTHERS

Su2023 **UCSC Science Internship Program (High School Students)**, AI Art Literature Survey with
 Lynn Dai, Adanur Durunas, Venkat Ramisetty, preliminary work for conference paper (Zhou
 et al. 2024, May)

SERVICE

DEPARTMENTAL SERVICE

2023-2024 **Community Manager**, UCSC Computational Media Department
 2020-2023 **Community Management Assistant**, UCSC Computational Media Department

OTHER SERVICES

2025 **Vice President**, Carleton University Postdoctoral Fellows Union

REVIEWING AND PROGRAM COMMITTEE

2026 **CHI Play**, Journal
 2026 **Leonardo**, Journal
 2026 **Digital Games Research Association (DiGRA)**, Conference
 2025-2026 **Foundation of Digital Games (FDG)**, Conference
 2025 **Conference on Human Factors in Computing Systems (CHI)**, Conference
 2023-2025 **International Conference on Interactive Digital Storytelling (ICIDS)**, Conference

SKILLS

GAME MAKING: Unity, Godot, Gamemaker, Construct 3

SOFTWARE ENGINEERING: React.js, P5.js, Shaders, Nvidia CUDA, OpenGL, C3.js, three.js, SQL, Google Magenta, PrettyMIDI, Python, C++, Clojure, Java

ARTISTIC & OTHERS: Cello, Piano, Figure Drawing, Digital Painting, Mandarin

REFERENCES

Michael Mateas (Ph.D. Advisor)

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Professor, Computational Media

University of California, Santa Cruz

Noah Wardrip-Fruin (Ph.D. Advisor)

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Professor, Computational Media

University of California, Santa Cruz

Edward Melcer (Postdoctoral Supervisor)

EdwardMelcer@cunet.carleton.ca

613-520-4333 (School of Computer Science)

Professor, Computer Science

Carleton University